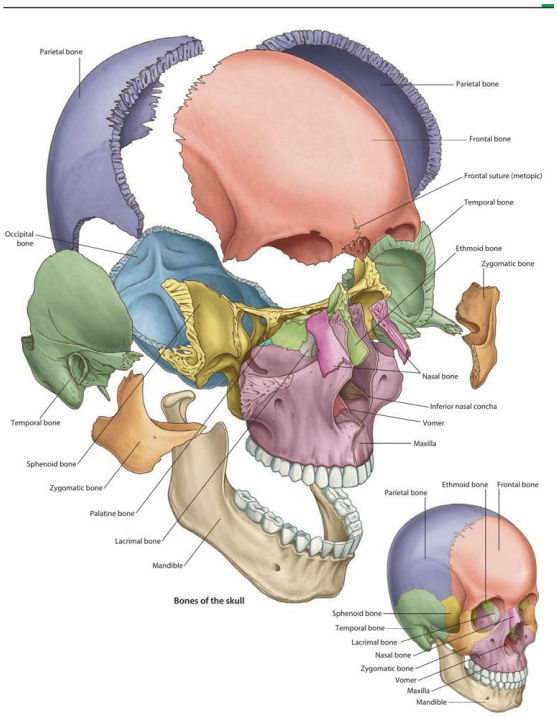
Skull bones

The dry skull for the practical: the views you are handed, the three cranial fossae, every foramen with its contents, and the individual bones.

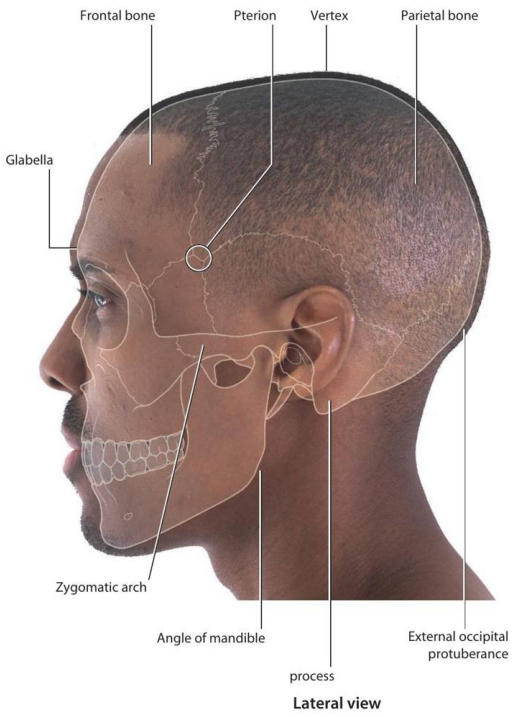
The 22 bones of the skull

- **Cranial bones (8):** frontal, ethmoid, sphenoid and occipital (single); parietal and temporal (paired).
- **Facial bones (14):** nasal, lacrimal, zygomatic, maxilla, palatine and inferior concha (paired); vomer and mandible (single).
- The **ear ossicles** (six in all) are usually counted separately. The **hyoid** is not part of the skull.
- Most skull bones meet at **sutures**. The **temporomandibular joint** is the skull's main synovial joint.

Clinical pearl: If you are asked how many bones are in the skull, state whether you are counting the ossicles.



Bones of the skull (exploded view). Gray's Atlas of Anatomy, 2nd ed., p. 477

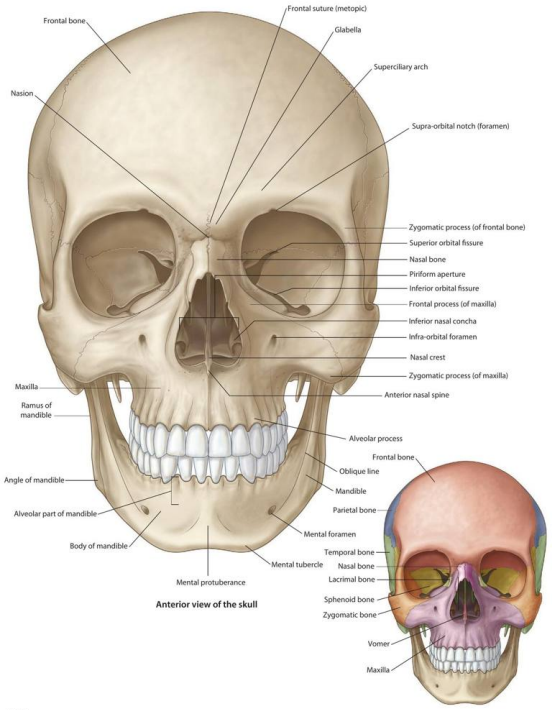


Surface anatomy with bones: lateral view (pterion, mastoid, angle). Gray's Atlas of Anatomy, 2nd ed., p. 476

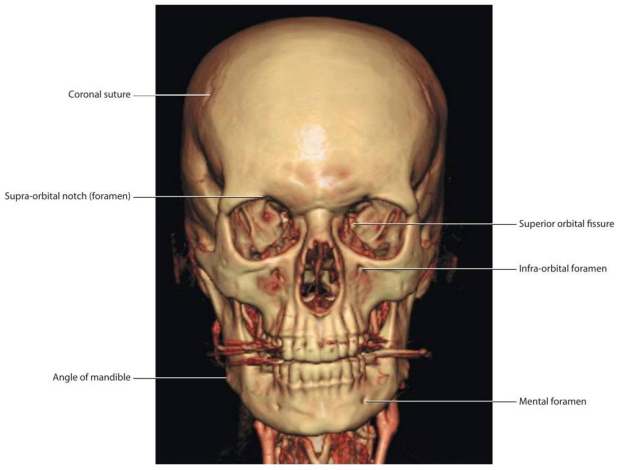
Norma frontalis: the skull from in front

- **Landmarks:** glabella, nasion, superciliary arch, supra-orbital margin, piriform aperture, anterior nasal spine.
- **Three foramina on the face, roughly in one vertical line: supra-orbital** (V1), **infra-orbital** (V2), **mental** (V3).
- **Orbital openings:** optic canal, superior and inferior orbital fissures.
- The **piriform aperture** is bounded by the nasal bones above and the maxillae at the sides and below.

Clinical pearl: The three facial foramina are where you can block each division of the trigeminal nerve.



Skull, anterior view: orbits, piriform aperture and maxilla. Gray's Atlas of Anatomy, 2nd ed., p. 478



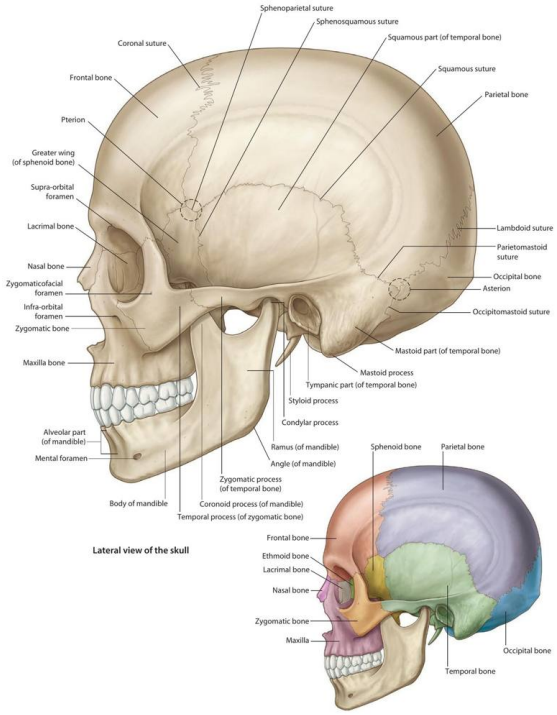
Anterior view of the skull.
Volume-rendered anterior view using multidetector computed tomography

Skull, anterior view: volume-rendered CT. Gray's Atlas of Anatomy, 2nd ed., p. 479

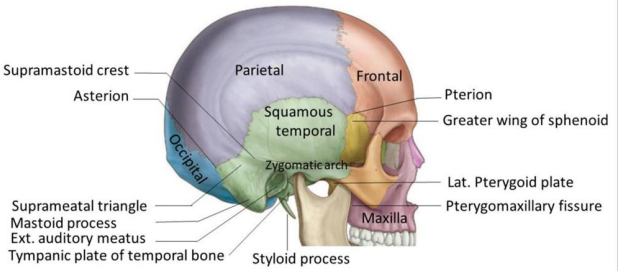
Norma lateralis: the skull from the side

- **Pterion:** an H-shaped junction of **four bones**: frontal, parietal, squamous temporal and greater wing of sphenoid. It lies about 4 cm above the zygomatic arch and about 3 cm behind the frontozygomatic suture.
- **Asterion:** where the **lambdoid, parietomastoid and occipitomastoid sutures** meet. It lies close to the junction of the transverse and sigmoid sinuses.
- **Zygomatic arch:** temporal process of the zygomatic bone plus zygomatic process of the temporal bone.
- **Also identify:** temporal lines, external acoustic meatus, mastoid and styloid processes, mandibular fossa and articular tubercle.

Clinical pearl: Pterion overlies the anterior division of the middle meningeal artery, the classic source of an extradural haematoma.



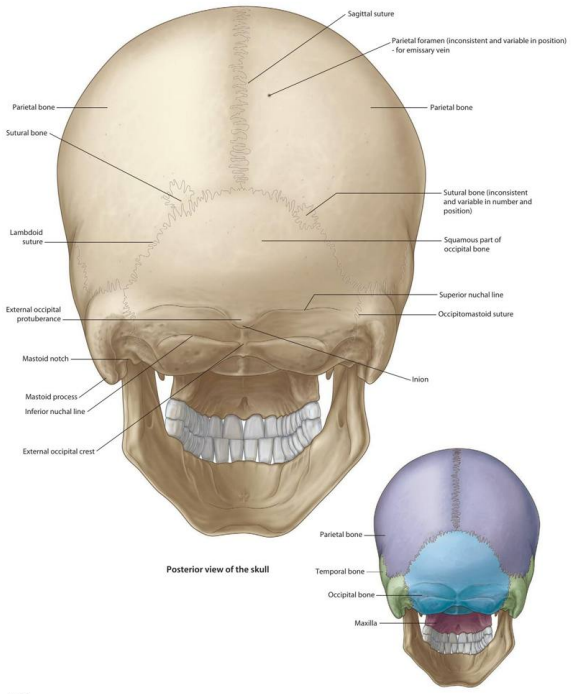
Lateral view of the skull: squamous, tympanic and mastoid parts. Gray's Atlas of Anatomy, 2nd ed., p. 480



Lateral skull: pterion, asterion and the infratemporal surface. Dr. Sameh Doss handout, p. 9

Norma occipitalis and verticalis

- **From behind:** lambdoid suture, external occipital protuberance (its midpoint is the inion), superior and inferior nuchal lines, external occipital crest, mastoid process and mastoid notch.
- **From above:** frontal, two parietals and occipital. The sutures are coronal, sagittal and lambdoid.
- **Bregma:** coronal meets sagittal, the site of the anterior fontanelle. **Lambda:** sagittal meets lambdoid, the site of the posterior fontanelle.
- **Sutural (Wormian) bones** are inconstant extra bones, most often in the lambdoid suture.

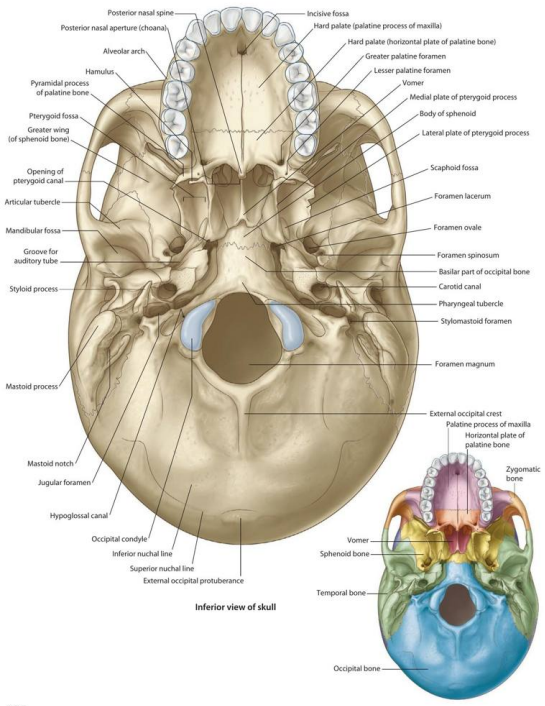


Posterior view of the skull: mastoid process and notch, occipital bone. Gray's Atlas of Anatomy, 2nd ed., p. 482

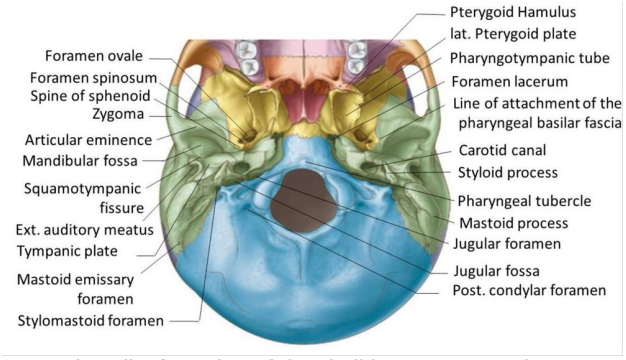
Norma basalis: the skull from below

- **Hard palate:** palatine process of the maxilla in front, horizontal plate of the palatine bone behind.
- **Palatal openings:** incisive fossa in the midline; greater palatine foramen near the third molar; lesser palatine foramina behind it.
- **Middle part:** medial and lateral pterygoid plates, hamulus, scaphoid fossa, foramen ovale, foramen spinosum, foramen lacerum, spine of sphenoid, groove for the auditory tube.
- **Posterior part:** carotid canal, jugular foramen, styloid process, stylomastoid foramen, mastoid process and notch, occipital condyle, hypoglossal canal, foramen magnum, pharyngeal tubercle.

Clinical pearl: The stylomastoid foramen lies between the styloid and mastoid processes. It is where the facial nerve leaves the skull.



Inferior view of the skull: foramina of the skull base. Gray's Atlas of Anatomy, 2nd ed., p. 484

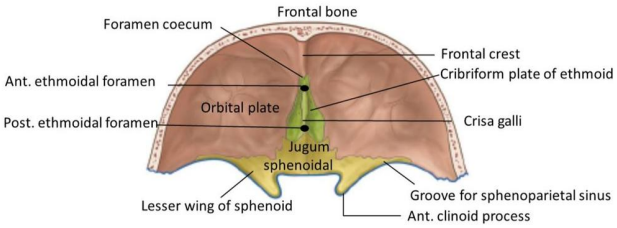


Norma basalis: foramina of the skull base. Dr. Sameh Doss handout, p. 16

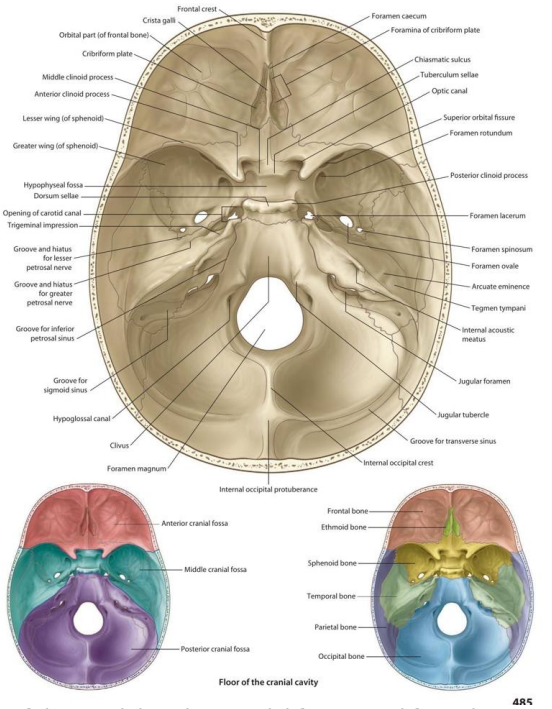
Anterior cranial fossa

- **Floor:** orbital part of the frontal bone, cribriform plate of the ethmoid, and the lesser wings and front of the body of the sphenoid.
- **Crista galli:** gives attachment to the falx cerebri.
- **Cribriform plate:** olfactory nerve filaments. The **anterior ethmoidal nerve and vessels** also enter the nasal cavity near its front.
- **Foramen caecum:** in front of the crista galli; may transmit a small emissary vein.

Clinical pearl: A fracture of the cribriform plate can cause CSF rhinorrhoea and anosmia.



Anterior cranial fossa. Dr. Sameh Doss handout, p. 24

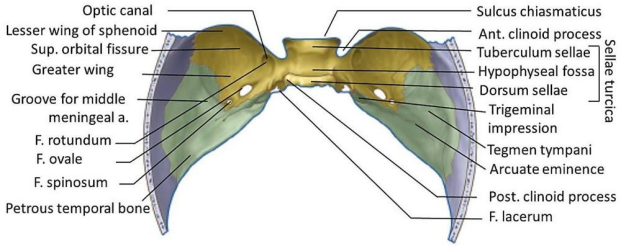


Floor of the cranial cavity: cranial fossae and foramina. Gray's Atlas of Anatomy, 2nd ed., p. 485

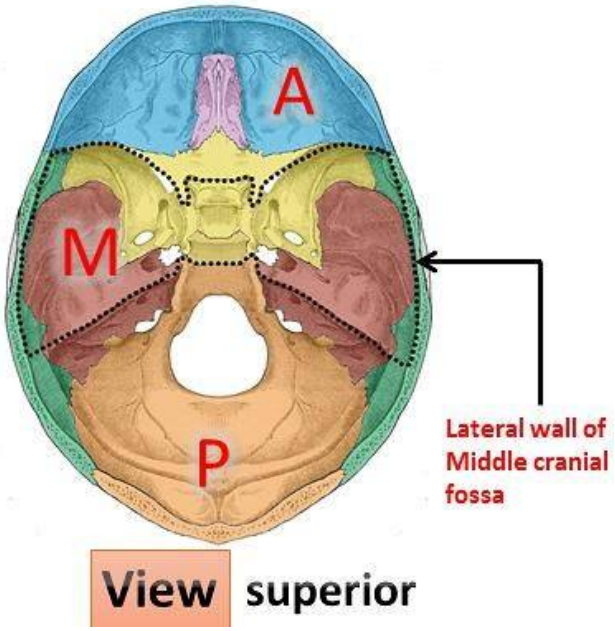
Middle cranial fossa

- **Floor:** greater wing and body of the sphenoid, plus the squamous and petrous parts of the temporal bone. The **sella turcica** lies in the centre.
- **Optic canal:** optic nerve and ophthalmic artery. **Superior orbital fissure:** III, IV, VI, lacrimal, frontal and nasociliary nerves, and the superior ophthalmic vein.
- **Foramen rotundum:** V2. **Foramen ovale:** V3, accessory meningeal artery, lesser petrosal nerve, emissary vein. **Foramen spinosum:** middle meningeal artery and the meningeal branch of V3.
- **Foramen lacerum:** closed by cartilage in life. The internal carotid artery crosses its upper part.
- **Petrous landmarks:** trigeminal impression, hiatus for the greater and lesser petrosal nerves, arcuate eminence, tegmen tympani.

Clinical pearl: The tegmen tympani is a thin plate of bone between the middle ear and the middle cranial fossa, so infection can spread through it.



Middle cranial fossa and its foramina. Dr. Sameh Doss handout, p. 25

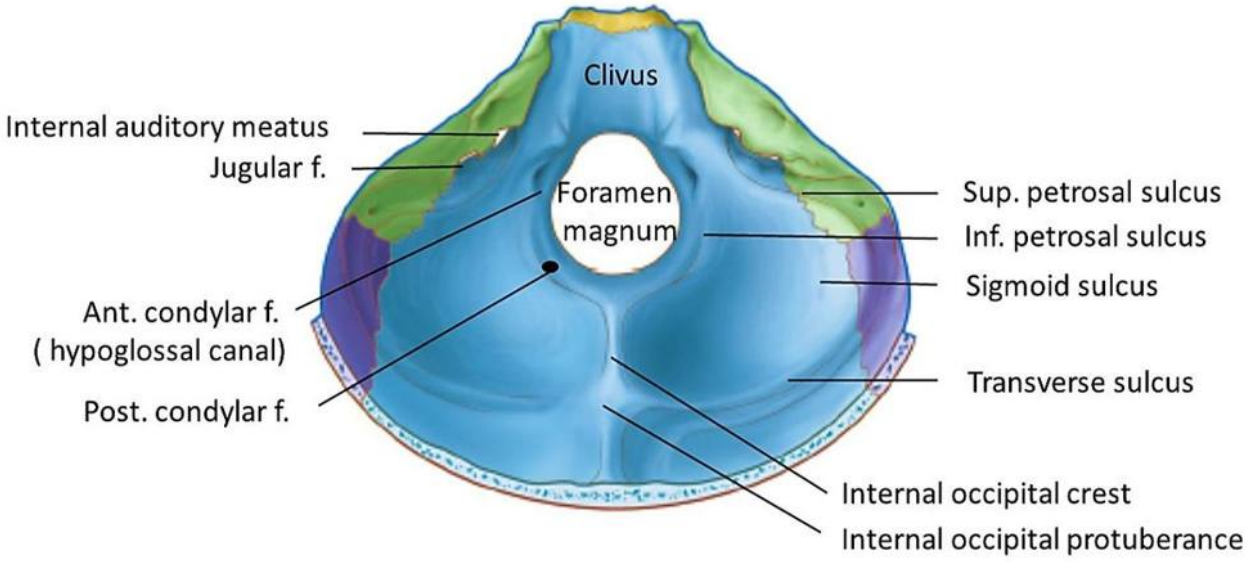


Cranial fossae from above: the petrous ridge divides middle from posterior fossa. ENT Steps Anatomy, p. 19

Posterior cranial fossa

- **Bones:** occipital, the back of the petrous and mastoid parts of the temporal, and a small part of the sphenoid (dorsum sellae and the clivus above the basi-occiput).
- **Internal acoustic meatus:** VII, VIII and the labyrinthine artery.
- **Jugular foramen:** IX, X, XI, the internal jugular vein (from the sigmoid sinus) and the inferior petrosal sinus.
- **Hypoglossal canal:** XII. **Foramen magnum:** medulla and meninges, vertebral arteries, spinal arteries and the spinal roots of XI.
- **Grooves:** transverse sinus, sigmoid sinus, inferior petrosal sinus.

Clinical pearl: A jugular foramen lesion affects IX, X and XI together (Vernet's syndrome).

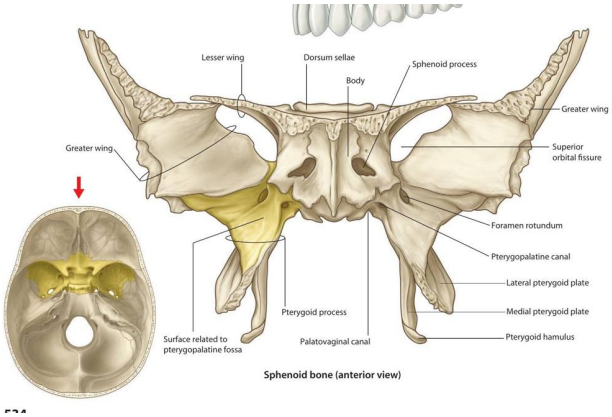


Posterior cranial fossa. Dr. Sameh Doss handout, p. 28

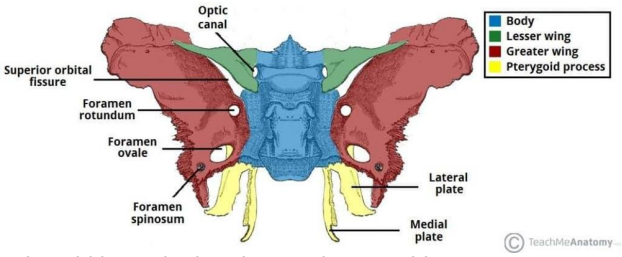
Sphenoid bone

- **Parts:** body (contains the sphenoidal sinus), two lesser wings, two greater wings, two pterygoid processes.
- **Greater wing foramina, front to back:** rotundum, ovale, spinosum.
- **Superior orbital fissure:** the gap between the lesser and greater wings. **Optic canal:** at the root of the lesser wing.
- **Pterygoid process:** medial plate ending in the hamulus, lateral plate, pterygoid fossa between them, scaphoid fossa at the top of the medial plate.
- **Pterygoid canal:** carries the nerve of the pterygoid canal (Vidian nerve) forward to the pterygopalatine fossa.

Clinical pearl: The tendon of tensor veli palatini turns around the hamulus to reach the palate.



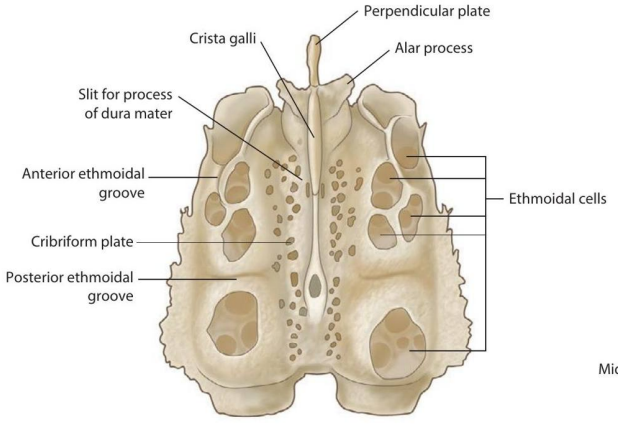
Sphenoid bone (anterior view). Gray's Atlas of Anatomy, 2nd ed., p. 534



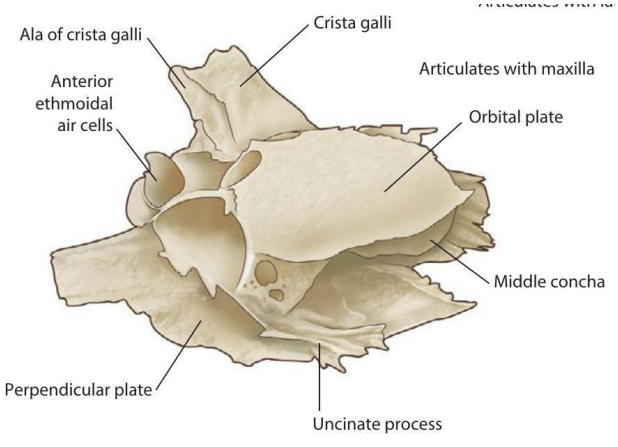
Sphenoid bone: body, wings and pterygoid processes. ENT Steps Anatomy, p. 147

Ethmoid, maxilla, palatine and small facial bones

- **Ethmoid:** cribriform plate with crista galli, perpendicular plate, and two labyrinths carrying the air cells, superior and middle conchae and uncinate process.
- **Maxilla:** a body containing the maxillary sinus, and **four processes:** frontal, zygomatic, alveolar and palatine.
- **Palatine bone:** L-shaped, with horizontal and perpendicular plates, a pyramidal process, and orbital and sphenoidal processes. The **sphenopalatine notch** between these last two forms the sphenopalatine foramen.
- **Nasal septum (bony part):** perpendicular plate of the ethmoid and the vomer.
- **Inferior concha** is a separate bone. The superior and middle conchae belong to the ethmoid.



Ethmoid bone (superior view): crista galli and cribriform plate. Gray's Atlas of Anatomy, 2nd ed., p. 486

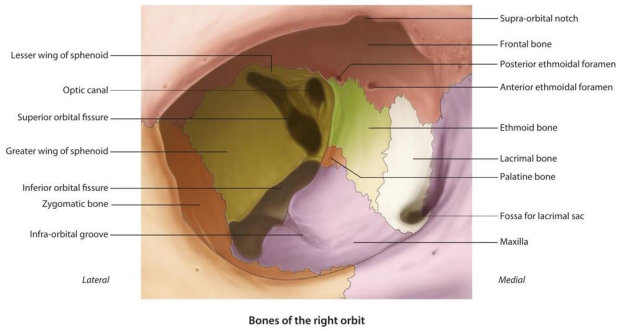


Ethmoid bone (lateral view): uncinate process, middle concha, orbital plate. Gray's Atlas of Anatomy, 2nd ed., p. 486

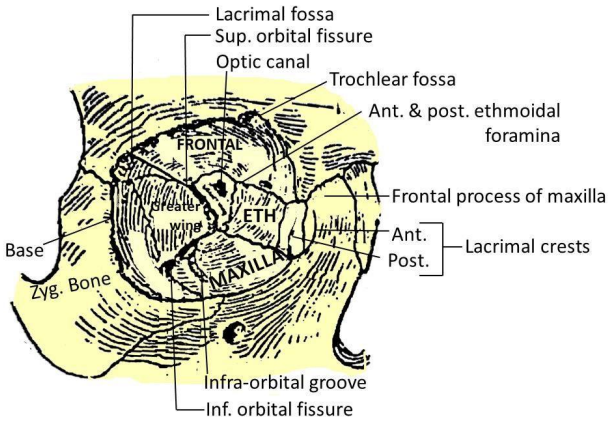
Bony orbit

- **Roof:** orbital plate of the frontal, lesser wing of sphenoid.
- **Medial wall:** frontal process of maxilla, lacrimal, orbital plate of ethmoid (lamina papyracea), body of sphenoid.
- **Floor:** maxilla, zygomatic and orbital process of the palatine.
- **Lateral wall:** zygomatic and greater wing of sphenoid.

Clinical pearl: The thin lamina papyracea and the floor over the infra-orbital canal are the usual sites of blow-out fracture.



Bones of the right orbit. Gray's Atlas of Anatomy, 2nd ed., p. 508

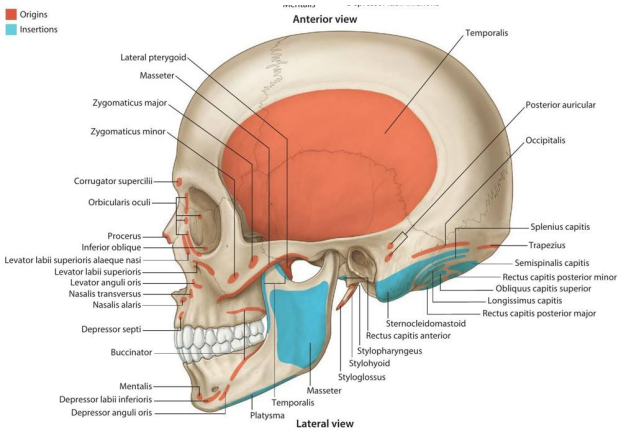


Bony orbit: walls, fissures and foramina. Dr. Sameh Doss handout, p. 90

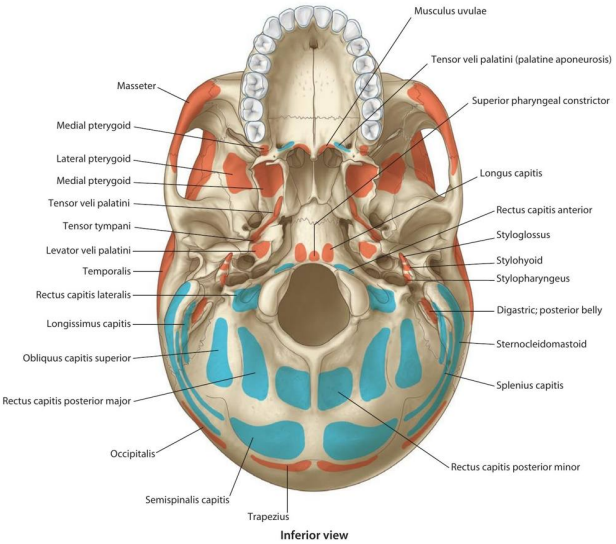
Muscle attachments on the skull

- **Temporal fossa:** temporalis. **Zygomatic arch:** masseter.
- **Mastoid process:** sternocleidomastoid, splenius capitis, longissimus capitis. **Mastoid notch:** posterior belly of digastric.
- **Styloid process:** styloglossus, stylohyoid, stylopharyngeus.
- **Pterygoid plates:** lateral pterygoid (lateral surface of the lateral plate), medial pterygoid (medial surface of the lateral plate).
- **Pharyngeal tubercle:** pharyngeal raphe. **Superior nuchal line:** trapezius and sternocleidomastoid.

Clinical pearl: Three styloid muscles have three different nerves: styloglossus XII, stylohyoid VII, stylopharyngeus IX.



Skull, lateral view: muscle attachments. Gray's Atlas of Anatomy, 2nd ed., p. 488



Skull, inferior view: muscle attachments. Gray's Atlas of Anatomy, 2nd ed., p. 489

Skull openings at a glance

Opening	Bone	Contents
Cribriform plate	Ethmoid	Olfactory nerve filaments
Optic canal	Lesser wing of sphenoid	Optic nerve, ophthalmic artery
Foramen rotundum	Greater wing of sphenoid	V2
Foramen spinosum	Greater wing of sphenoid	Middle meningeal artery
Internal acoustic meatus	Petrous temporal	VII, VIII, labyrinthine artery
Stylomastoid foramen	Temporal	Facial nerve
Hypoglossal canal	Occipital	XII
Infra-orbital foramen	Maxilla	Infra-orbital nerve (V2)
Greater palatine foramen	Palatine and maxilla	Greater palatine nerve and vessels

Notes are a study summary of standard anatomy. Check them against your course material. Personal study use.